

- 29** The rest mass of the deuterium, ${}^2_1\text{H}$ is equivalent to an energy of 1876 MeV. The rest mass of a proton is equivalent to 939 MeV and that of a neutron to 940 MeV.

How may a deuterium disintegrate into a proton and a neutron?

- A** by capturing a γ -ray photon of energy 2 MeV
- B** by emitting a γ -ray photon of energy 2 MeV
- C** by capturing a γ -ray photon of energy 3 MeV
- D** by emitting a γ -ray photon of energy 3 MeV

29	C	Mass defect in terms of energy = $939 + 940 - 1876 = 3 \text{ MeV}$ 3 MeV needs to be supplied to break up the nucleus
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[Turn over

20 Samples of two radioactive nuclides, X and Y , each have equal activity, A_0 , at time $t = 0$. X